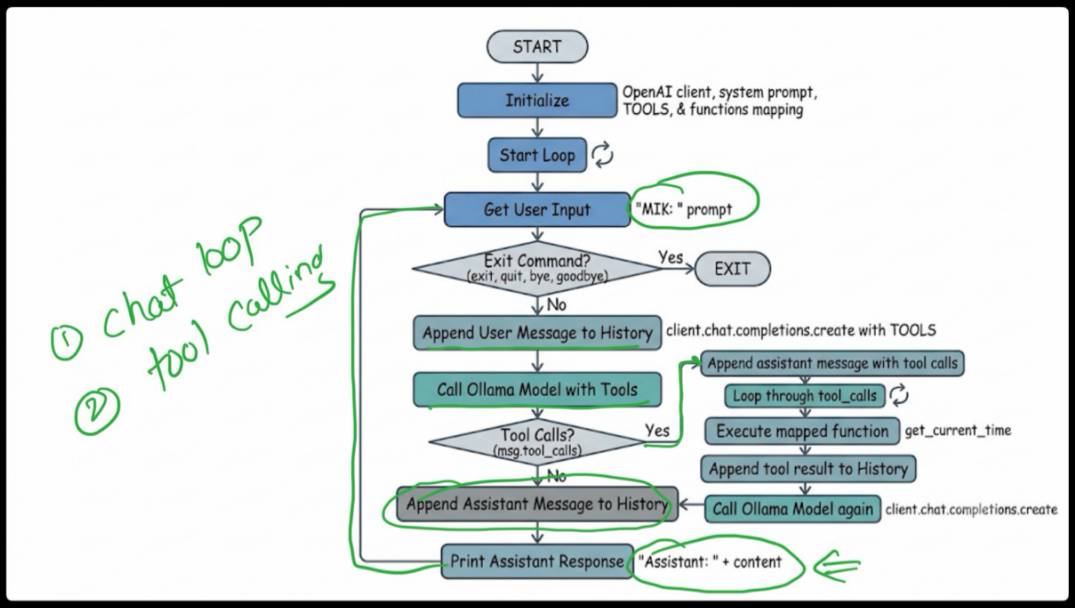


codestorywithmiK

CSwithMIK

codestorywithMIK

Recap : video - 7



tools.py > ...

```python
from datetime import datetime

def get_current_time():
    return datetime.now().strftime("%Y-%m-%d %H:%M:%S")


#TOOLS = [{}, {}, {}]
TOOLS = [
    {
        "type" : "function",
        "function" : {
            "name" : "get_current_time",
            "description" : "Returns the current date and time as a formatted string.",
            "parameters" : { "type" : "object", "properties" : {}},
        },
    }
]


FUNCTION_NAME_TO_FUNCTION_MAP = {
    "get_current_time" : get_current_time,
}
```

get_current_time() has no parameters

Let's add a tool with parameter.

→ Structure of one tool

```
{
    "type": "function",                 # always "function"
    "function": {
        "name": "function_name",        # name the model will call
        "description": "what it does and when to use it",
        "parameters": { ... },          # JSON Schema describing the arguments
    },
}
```

Format for parameters

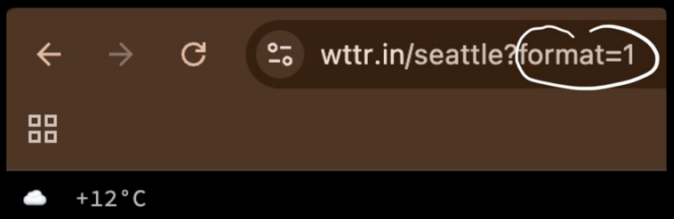
```
"parameters": {
    "type": "object",                    # always "object" at the top level
    "properties": {                      # one entry per argument
        "argument_name_1": {
            "type": "string",
            "description": "what this argument is",
        },
        "argument_name_2": {
            "type": "string",
            "enum": ["option_a", "option_b"],      # optional: restrict to fixed choices
            "description": "what this argument is",
        },
        "argument_name_3": {
            "type": "array",
            "items": {"type": "string"},           # type of each element
            "description": "what this argument is",
        },
    },
    "required": ["argument_name_1"],     # list of mandatory arguments
}
```

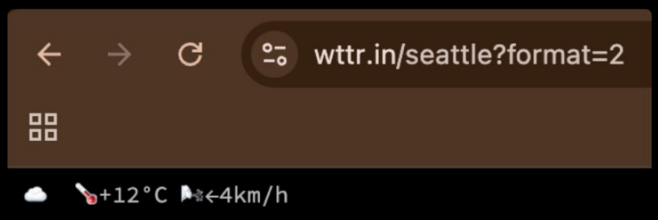
Tool : get_city_weather

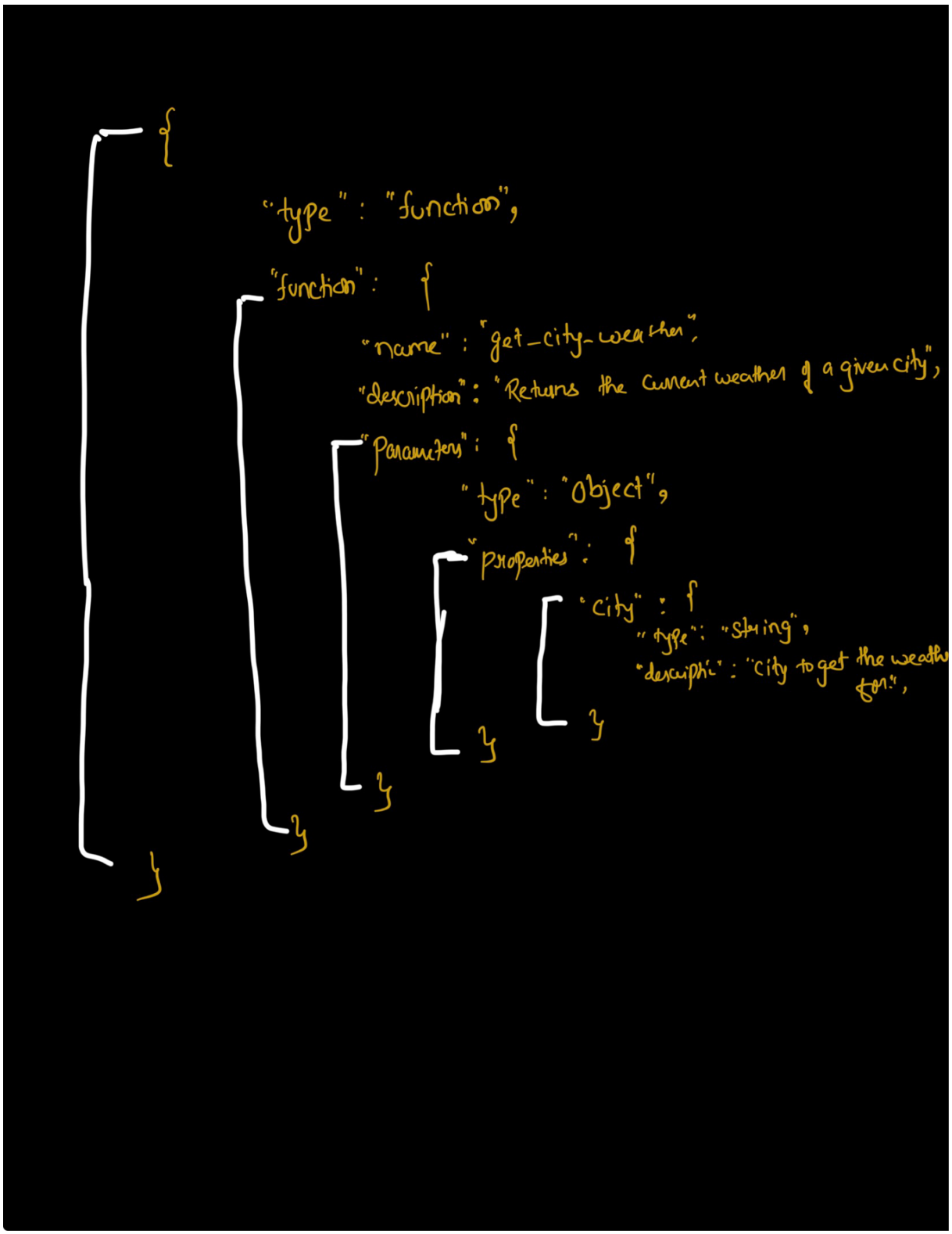
```
def get_city_weather(city):
    url = f"https://wttr.in/{city}?format=3"
    resp = requests.get(url, timeout=10)
    return resp.text
```

https://wttr.in → Free weather service (web API)

https://wttr.in/seattle?format=3






{
"type" : "function",
"function" : {
"name" : "get_city_weather",
"description" : "Returns the current weather of a given city",
"parameters" : {
"type" : "object",
"properties" : {
"city" : {
"type": "string",
"descriptic" : "city to get the weathe
for",
}
}
}
}
}

pip install requests

response = requests(URL, timeout=10)

return response.text

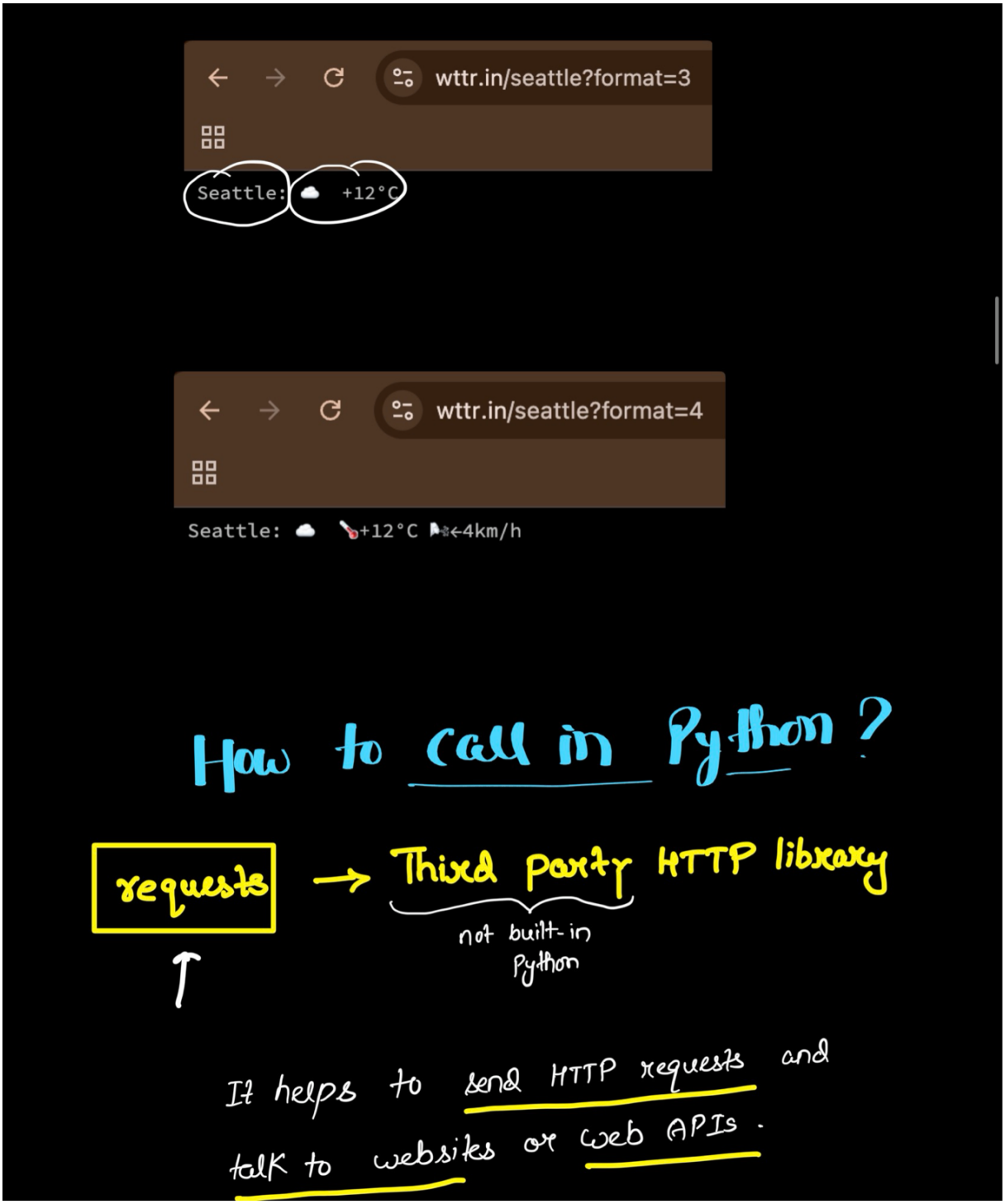
tools.py > ...

```python
from datetime import datetime

def get_current_time():
    return datetime.now().strftime("%Y-%m-%d %H:%M:%S")


#TOOLS = [{}, {}, {}]
TOOLS = [
    {
        "type" : "function",
        "function" : {
            "name" : "get_current_time",
            "description" : "Returns the current date and time as a formatted string.",
            "parameters" : { "type" : "object", "properties" : {}},
        },
    }
]


FUNCTION_NAME_TO_FUNCTION_MAP = {
    "get_current_time" : get_current_time,
    "get_city_weather" : get_city_weather,
}
```


wttr.in/seattle?format=3
Seattle: +12°C
wttr.in/seattle?format=4
Seattle: +12°C ←4km/h
How to call in Python?
requests
Third party HTTP library
not built-in Python
It helps to send HTTP requests and talk to websites or web APIs.